

## **DEVICE AUTHENTICATION, SECURITY, AND DATA PROTECTION EMBODIMENTS**

In various embodiments, the intraluminal compliance and distance profiling device incorporates authentication, cybersecurity, and data protection mechanisms configured to safeguard sensitive physiologic information and prevent unauthorized device operation or data access.

The device may include secure identity elements such as embedded cryptographic chips, secure microcontrollers, trusted execution environments, or hardware-based authentication modules configured to verify device authenticity during communication with associated software systems.

In certain embodiments, communication between the device, companion applications, and remote computational platforms is protected using encrypted communication protocols. Encryption methods may include symmetric encryption, asymmetric encryption, secure key exchange procedures, or future cryptographic standards designed to protect transmitted measurement data.

Authentication procedures may verify authorized pairing between hardware devices and user accounts. Pairing mechanisms may include digital certificates, one-time authentication codes, near-field communication pairing, biometric confirmation, or secure provisioning workflows performed during initial device activation.

The system may further implement secure boot processes preventing execution of unauthorized firmware. Firmware images may be cryptographically signed and validated prior to installation, thereby preventing malicious modification or counterfeit firmware deployment.

In certain implementations, consumables, accessories, or replaceable modules may include machine-readable identifiers enabling verification of approved components. Authentication systems may prevent operation using unauthorized or counterfeit components while preserving user safety and measurement accuracy.

Data protection embodiments may include local encryption of stored measurement data, anonymization of transmitted datasets, role-based access control, and user-controlled permission frameworks governing data sharing with clinicians, researchers, or other authorized parties.

The computational ecosystem may support privacy-preserving computation techniques including secure aggregation, encrypted profile comparison, differential privacy methods, or anonymized feature representation exchange to minimize exposure of raw physiologic measurements.

In certain embodiments, security monitoring systems detect abnormal communication behavior, unauthorized access attempts, or anomalous firmware states. Detection events may trigger alerts, restricted operation modes, or automatic isolation from network communication until security verification is completed.

Accordingly, device authentication and data protection embodiments establish a secure operational foundation enabling trusted deployment of physiologic sensing systems within consumer, research, and clinical-support environments while maintaining confidentiality, integrity, and availability of sensitive data.